

Stabilizing unstable steady states using extended time-delay autosynchronization

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We describe a method for stabilizing unstable steady states in nonlinear dynamical systems using a form of extended time-delay autosynchronization. Specifically, stabilization is achieved by applying a feedback signal generated by high-pass-filtering in real time the dynamical state of the system to an accessible system parameter or variables. Our technique is easy to implement, does not require knowledge of the unstable steady state coordinates in phase space, automatically tracks changes in the system parameters, and is more robust to broadband noise than previous schemes. We demonstrate the controller's efficacy by stabilizing unstable steady states in an electronic circuit exhibiting low-dimensional temporal chaos. The simplicity and robustness of the scheme suggests that it is ideally suited for stabilizing unstable steady states in ultra-high-speed systems. © 1998 American Institute of Physics. [S1054-1500(98)00704-6]

In this article, we investigate a control scheme that is effective in suppressing deterministic chaos in dynamical systems. It is desirable to devise such schemes because the presence of deterministic chaos in devices generally degrades their performance in many applications. The signatures of chaos include erratic, noise-like fluctuations in the temporal evolution of the system variables, broadband features in the power spectrum, and the long-term behavior of the system is extremely sensitivity to applications of small perturbations to the system variables. Recent research has demonstrated that feedback control algorithms that take advantage of the special properties of chaotic systems are effective in controlling chaos. We describe a method for stabilizing the unstable steady states of dynamical systems that uses continuous feedback of an error signal generated by high-pass-filtering in real time the state of the system. The scheme is easy to implement and does not suffer from high-frequency instabilities occurring in other methods. We present experimental results of the control of a chaotic electronic circuit and find good agreement with theoretical predictions. We also note that an all-optical implementation of the controller is possible, suggesting that ultra-high-speed optical instabilities can be stabilized using this technique.

I. INTRODUCTION

In many cases of practical interest, the performance of nonlinear devices is limited by temporal fluctuations in the state variables of the device. One source of fluctuations is chaotic instabilities arising from the inherent nonlinear dynamical behavior of the device, a particularly insidious source of erratic dynamics since it cannot be suppressed by

better shielding from the environment. For example, it is desirable that the intensity of the semiconductor laser in a compact disc player remains constant as a function of time. However, weak optical feedback from the disc can induce chaotic instabilities that limit the ability of the player to detect information stored on the disc.¹ One solution to this problem is to redesign the entire system to avoid chaos, but this may prove impractical or costly.

We present a more subtle solution for stabilizing the unstable steady states (USS's) of nonlinear systems based on a controlling-chaos scheme known as extended time-delay autosynchronization (ETDAS).²⁻⁵ Specifically, the control protocol attempts to stabilize an USS by making small adjustments $\epsilon(t)$ to an accessible system parameter q (nominal value $\bar{q}$) when the system is in a neighborhood of the state. The scheme uses a continuous-time feedback loop in which the error signal $\epsilon(t)$ is generated in real time by high-pass-filtering the dynamical state of the system $\xi(t) = \hat{n} \cdot \mathbf{z}(t)$, where $\mathbf{z}(t)$ is the system state vector and $\hat{n}$ is the measurement direction in phase space. Our scheme is easy to implement and does not require knowledge of the coordinate $\mathbf{z}_*$ of the desired USS. In addition, an all-optical implementation of the controller is possible,⁶ suggesting that ultra-high-speed optical instabilities can be stabilized using this technique. We demonstrate the feasibility of the scheme by stabilizing USS's of a nonlinear electronic circuit operating in the chaotic and nonchaotic regimes.

The possibility that feedback can efficiently stabilize unstable states embedded within a chaotic system, such as USS's and unstable periodic orbits (UPO's), was recently put forth by Ott, Grebogi, and Yorke (OGY).⁷ They proposed that variants of standard modern control engineering techniques can be used to stabilize the desired unstable state without using a detailed model of the system.⁸ In their original conceptualization, the state of the system is sensed and adjustments are made to the accessible system parameter as

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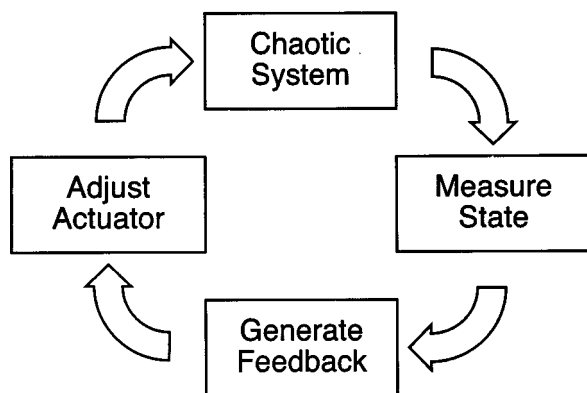


FIG. 1. Generalized block diagram for feedback control of chaos.

the system passes through a surface of section, where the size of the adjustments is proportional to the difference between the current and desired states. Their technique is easy to implement and has the advantage that the control parameters can be determined straightforwardly from experimental observations. Several research groups have used the OGY idea to control chaos experimentally in mechanical, optical, electrical, and biological systems.⁹

In general, feedback control of chaotic systems involves measuring the state $\xi(t)$ of the system, generating an appropriate feedback signal, and adjusting an actuator that modifies an accessible system parameter by an amount $\epsilon(t)$, as shown schematically in Fig. 1. Feedback schemes can stabilize the system using only small perturbations due to the sensitivity of chaotic systems to initial conditions. In addition, the size of the feedback signal is small and little energy is expended by the controller when the system is stabilized because the scheme stabilizes a state that is embedded within the chaotic system.

While most research on controlling chaos has focused on stabilizing UPO's, a few research groups have investigated schemes for the efficient control of USS's. Roy *et al.*¹⁰ and Johnson and Hunt¹¹ have found that occasional proportional feedback (OPF) can stabilize the USS's of a laser and an electronic circuit, respectively. The OPF scheme¹² is a variant of the original OGY control method, where the OPF controller generates a series of discrete fixed-duration "kicks" of magnitude $\epsilon(t)$ to an accessible system parameter that directs the system towards the desired state by comparing the current state $\xi(t)$ of the system to an empirically generated reference state. Carr and Schwartz¹³ have found that using the duration of the perturbations as an additional feedback parameter makes it possible to stabilize USS's embedded in high-dimensional systems. We note that controllers based on neural networks,¹⁴ continuous proportional feedback,¹⁵ nonlinear continuous feedback,¹⁶ Kalman filtering techniques,¹⁷ and map-based techniques¹⁸ have also been considered.

In a subsequent investigation of the OPF feedback scheme, Johnson and Hunt¹⁹ determined that the only part of the feedback signal $\epsilon(t)$ responsible for control was proportional to the derivative of $\xi(t)$. In a separate set of experiments, Bielawski *et al.*²⁰ stabilized the dynamics of a fiber

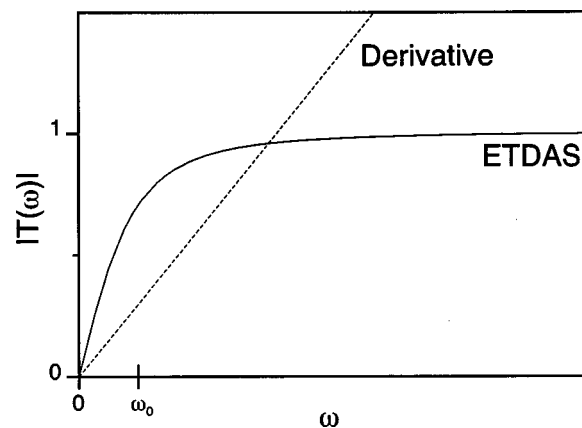


FIG. 2. The transfer functions of the derivative control scheme (dashed line) and the form of ETDAS feedback optimized for the control of USS's (solid line).

laser, and Parmananda *et al.*²¹ controlled an electrochemical cell using a feedback signal of the form

$$\epsilon(t) = -\gamma \frac{d\xi(t)}{dt}, \quad (1)$$

where γ is the feedback gain and $\xi(t)$ was taken to be the laser intensity. We note that modern control engineers use derivative control to stabilize nonchaotic systems.

Derivative control has several key advantages over previous schemes for controlling USS's: it eliminates the need for a reference state (related to a projection of $\mathbf{z}_*$) with which to compare the system's current behavior, and it offers the possibility of stabilizing highly unstable states since it generates a continuous feedback signal. Despite these advantages, derivative control is not perfect in an experimental setting because it can accentuate fluctuations that are often unrelated to the unstable behavior inherent to the nonlinear system. To understand the origin of the noise sensitivity, it is useful to consider the effects of the controller in the frequency domain. A frequency-domain analysis is possible because the feedback signal is linearly proportional to the input signal and hence $\epsilon(\omega) = -\gamma T(\omega)\xi(\omega)$, where $\xi(\omega)$ and $\epsilon(\omega)$ are the spectra of the input and output signals, respectively, $T(\omega)$ is the transfer function of the controller, and γ is the feedback gain. For feedback of the form of Eq. (1), $T(\omega) = i\omega$; we show $|T(\omega)|$ as the dotted line in Fig. 2. It is apparent from the figure that derivative control will be very sensitive to high-frequency information contained in $\xi(\omega)$ since $|T(\omega)| \rightarrow \infty$ as $\omega \rightarrow \infty$. Since the spectrum of the chaotic fluctuations is often band limited, the controller may feedback a high-frequency signal arising from random fluctuations (noise) in the system rather than the chaotic fluctuations. This effect can lead to high-frequency instabilities and large power dissipation in the feedback loop.

II. CONTROLLING UNSTABLE STEADY STATES USING ETDAS

We propose a control scheme that efficiently stabilizes USS's while avoiding the disadvantages of derivative control. Our scheme is motivated by a method for controlling

UPO's called extended time-delay autosynchronization (ETDAS).²⁻⁵ This method stabilizes UPO's of period τ by application of a continuous feedback signal that is proportional to the difference between the current state of the system and an infinite series of values of the state time-delayed by integral multiples of τ . Specifically, the dynamics of an ETDAS controller are governed by

$$\epsilon(t) = -\gamma \left[\xi(t) - (1-R) \sum_{k=1}^{\infty} R^{k-1} \xi(t-k\tau) \right], \quad (2)$$

where $\epsilon(t)$ is the feedback signal, $\xi(t)$ is the measured state of the system, γ is a feedback gain parameter, and $-1 \leq R < 1$ regulates the weight of time-delayed information. The case when $R=0$ corresponds to time-delay autosynchronization (TDAS) in which the feedback signal is based on a single previous state of a chaotic system.²² Note that $\epsilon(t)$ vanishes for any value of R when control is successful since $\xi(t) = \xi(t-k\tau)$ for all k . Feedback of the form given by Eq. (2) has been used to stabilize UPO's of a high-speed electronic circuit and has been shown to stabilize certain traveling waves states in systems that display spatio-temporal complexity.^{23,24} In addition, it can be used to stabilize USS's for an appropriate choice of τ with $R=0$.²⁵

We show that USS's can be stabilized efficiently using a form of Eq. (2) with an appropriate choice of the feedback parameters; a frequency domain analysis of the ETDAS scheme gives us some guidance concerning this choice. Note that the feedback signal given by Eq. (2) linearly relates the input $\xi(t)$ with the output signal $\epsilon(t)$ so that $\epsilon(\omega) = -\gamma T(\omega) \xi(\omega)$, where

$$T(\omega) = \frac{1 - e^{i\omega\tau}}{1 - R e^{i\omega\tau}} \quad (3)$$

is the transfer function for ETDAS feedback. The transfer function "filters" the observed state of the dynamical system, characterized by $\xi(\omega)$, to produce the necessary feedback signal.

Figure 3 shows the frequency dependence of $|T(\omega)|$ for $R=0.1$ and 0.8 , where it is seen that there are a series of "notches" that go to zero at multiples of the characteristic frequency $\omega_c = 2\pi/\tau$ of the desired UPO, and that the notches become narrower for larger R . The existence of the notches in $T(\omega)$ is necessary since $\epsilon(t)$, and hence $\epsilon(\omega)$, must vanish when the UPO is stabilized. Recall that the spectrum $\xi(\omega)$ of the system consists, in general, of a series of δ -functions at multiples of the characteristic frequency ω_c of the orbit when stabilized to the desired UPO; the filter must remove these frequencies (via the notches) so that $\epsilon(\omega) = 0$.

Based on this reasoning, an appropriate transfer function for stabilizing USS's consists of a notch at $\omega=0$ since the spectrum of an USS is a single δ -function at $\omega=0$. All other frequencies are passed by the filter with equal weight, thereby generating a negative feedback signal from this information. We can obtain this transfer function by decreasing the value of τ , thereby increasing the distance between notches, while simultaneously increasing the value of R in order to keep the controller sensitive to frequencies near ω

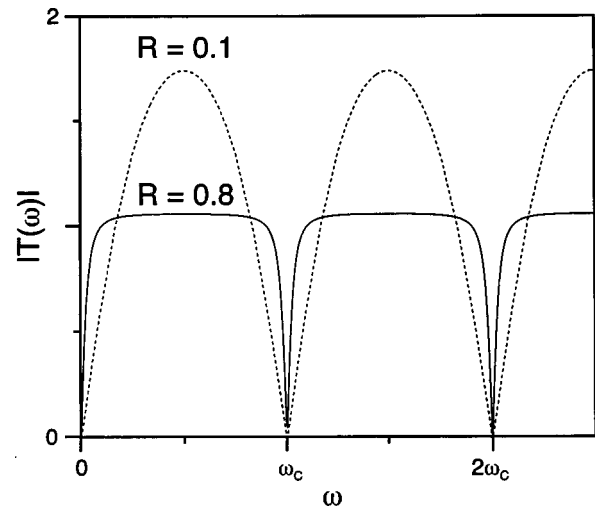


FIG. 3. Transfer function of the ETDAS feedback scheme for different values of parameter R . The feedback is more effective in stabilizing UPO's when R is closer to one because the controller is more sensitive to frequencies that could destabilize the orbit without generating destabilizing perturbations at other frequencies.

$=0$. Placing this discussion on a rigorous foundation, we rewrite Eq. (2) in an equivalent, recursive form as

$$\epsilon(t) = -\gamma [\xi(t) - \xi(t-\tau)] + R \epsilon(t-\tau), \quad (4)$$

and perform the limits $\tau \rightarrow 0$ and $R \rightarrow 1$ with $(1-R)/\tau \equiv \omega_0$ held finite, thereby obtaining a new feedback relation given by

$$\frac{d\epsilon(t)}{dt} = -\omega_0 \epsilon(t) - \gamma \frac{d\xi(t)}{dt}, \quad (5)$$

which is effective for stabilizing USS's. The feedback signal prescribed by Eq. (5) is identical to a single-pole high-pass filter familiar from elementary electronics. We note that ETDAS feedback in the form of Eq. (2) with τ small but finite and R near to but less than 1 does not necessarily stabilize USS's.

The feedback scheme given by Eq. (5) falls into a class of controlling chaos techniques that do not require a reference state (i.e., the coordinates of the USS or explicit knowledge of the UPO). This is advantageous in situations where it is difficult to obtain an accurate reference state such as in high-speed dynamical systems, or in biological systems where the dynamics tend to be nonstationary and the measurements are noisy. However, it may lead to complications when the system possesses more than one unstable state with similar properties since the control scheme cannot explicitly select one stable state from the others. The selection of specific states may be possible if the basin of attraction for the states in the presence of control are distinct, or if the domain of feedback parameters that successfully stabilize the USS's are distinct.

III. STABILITY ANALYSIS

The stability of an arbitrary USS in an N -dimensional phase space in the presence of feedback of the form given by

Eq. (5) can be determined straightforwardly using standard stability analysis techniques. Consider a system whose dynamical evolution is governed by

$$\frac{d\mathbf{z}}{dt} = \mathbf{F}(\mathbf{z}, q), \quad (6)$$

possessing an USS $\mathbf{z}_*$, where $\mathbf{z}$ is the N -dimensional state vector, $\mathbf{F}$ is the flow, and $q = \bar{q} + \epsilon(t)$ is an accessible system parameter. We introduce a new variable $\mathbf{x} \equiv \mathbf{z} - \mathbf{z}_*$ and linearize the dynamics about the USS so that

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{B}\epsilon, \quad (7)$$

where $\mathbf{A} = \partial\mathbf{F}/\partial\mathbf{x}|_{q=\bar{q}}$ is the Jacobian of the system in the absence of control, and $\mathbf{B} = \partial\mathbf{F}/\partial q|_{q=\bar{q}}$ quantifies the manner by which the control signal affects the state of the system. The combined dynamics of the system and the controller can be stated succinctly as

$$\frac{d\mathbf{y}}{dt} = \mathbf{G}\mathbf{y}, \quad (8)$$

where the $(N+1)$ -dimensional expanded state vector is given by

$$\mathbf{y} = \begin{bmatrix} \mathbf{x} \\ \epsilon \end{bmatrix}, \quad (9)$$

and the $(N+1) \times (N+1)$ matrix by

$$\mathbf{G} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ -\gamma\hat{\mathbf{n}}^T\mathbf{A} & -(\omega_0 + \gamma\hat{\mathbf{n}}^T\mathbf{B}) \end{bmatrix}. \quad (10)$$

The system (8) is stable if and only if $\mathbf{G}$ possesses eigenvalues whose real parts are negative. The values of the control parameters γ , ω_0 , and $\mathbf{B}$ rendering the system stable can be determined by calculating explicitly the eigenvalues of the matrix or by using the Routh–Hurwitz stability criterion²⁶ that does not require an explicit calculation of the eigenvalues of $\mathbf{G}$. Briefly, the Routh–Hurwitz criterion imposes constraints on the values of the characteristic polynomial coefficients of $\mathbf{G}$.

While predictions regarding the stability of an USS under feedback control requires a detailed analysis of the eigenvalues or characteristic polynomial, it is possible to make some general statements regarding the inability of the controller to stabilize certain classes of USS's. Based on a generalization of an analysis by Ushio,²⁷ it can be shown that feedback of the form given by Eq. (5) fails to control a USS for the case when the associated matrix $\mathbf{A}$ has an odd number of real eigenvalues whose size is greater than zero. The proof of this statement is given in the Appendix.

Finally, we note that Eq. (8) can be written in a slightly different form in which $\mathbf{G} = \tilde{\mathbf{A}} + \tilde{\mathbf{B}}\tilde{\mathbf{K}}^T$,⁸ where the matrix $\tilde{\mathbf{K}}^T$ has only terms proportional to the feedback parameters γ and ω_0 . This decomposition facilitates the use of modern control engineering techniques such as pole-placement and tests of controllability and observability.²⁸ We do not pursue this approach since, in general, these techniques require access to

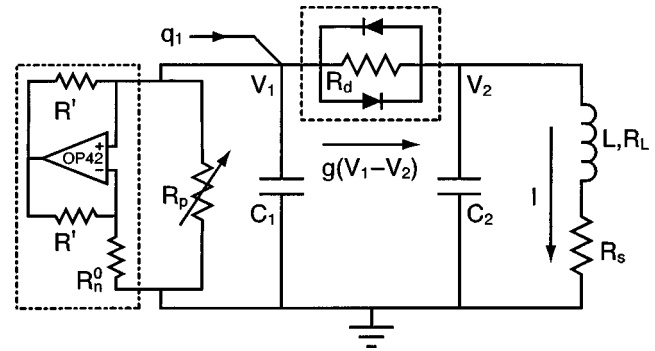


FIG. 4. Schematic diagram of the chaotic electronic oscillator. The collection of components in the dot-dashed box form a negative resistor (active device of resistance $-R_n^0$ with $R' = 300 \Omega$), and those in the dashed box form the nonlinear coupling described by the function g .

all dynamical variables or that they be reconstructed using mathematical models which is not in the spirit of most controlling chaos research.

IV. THE CHAOTIC SYSTEM

To demonstrate the efficacy of our method, we control the dynamics of a chaotic electronic circuit consisting of passive linear components, nonlinear signal diodes, and an active negative resistor,²⁹ as shown schematically in Fig. 4. The dynamics of the circuit are governed by the set of equations

$$C_1 \frac{dV_1}{dt} = \frac{V_1}{R_n} - g(V_1 - V_2) + q_1, \quad (11a)$$

$$C_2 \frac{dV_2}{dt} = g(V_1 - V_2) - I + q_2, \quad (11b)$$

$$L \frac{dI}{dt} = V_2 - R_m I + q_3, \quad (11c)$$

where V_1 (V_2) is the voltage drop across capacitor C_1 ($C_2 = 10$ nF), and I is the current flowing through the inductor $L = 55$ mH. The other circuit parameters are measured experimentally with 1% accuracy and are given by $1/R_n = 1/R_n^0 - 1/R_p$, $R_n^0 = 2.61$ k Ω is the maximum attainable value of the negative resistance, R_p is a ten-turn potentiometer with maximum resistance of 100 k Ω , $R_m = R_L + R_s = 455 \Omega$, $R_L = 355 \Omega$ is the dc resistance of the inductor, $g(V) = V/R_d + 2I_r \sinh(\alpha V/V_d)$ is the current flowing through the parallel combination of the resistor $R_d = 7.86$ k Ω and signal diodes (type 1N914), $I_r = 5.63$ nA is the reverse current of the diodes, $V_d = 0.58$ V is the diode voltage, and $\alpha = 11.6$. The parameter q_1 (q_2) represents the possibility of injecting a bias current into the V_1 -node (V_2 -node) of the circuit whose value is nominally equal to zero in the experiments, and q_3 represents the possibility of applying a bias voltage (nominal value equal to zero) across the inductor.

In some of the later sections, the discussion is greatly simplified if we express Eqs. (11) in dimensionless coordinates. Normalizing all voltages to V_d , resistances to R_c

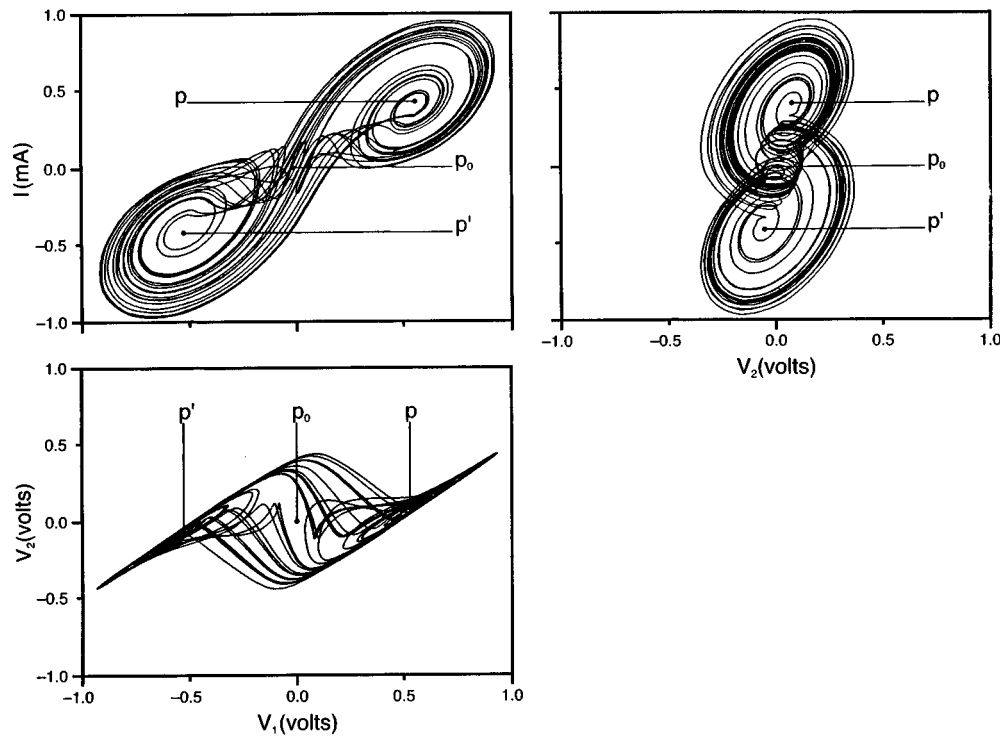


FIG. 5. Experimentally measured projections of the strange attractor for chaotic system in the “double-scroll” regime with $R_p = 20.0 \text{ k}\Omega$. The three fixed points embedded in the attractor are indicated as p , p' , and p_0 .

$=\sqrt{L/C_1} = 2.35 \text{ k}\Omega$, currents to $I_c = V_d/R_c = 0.25 \text{ mA}$, and time to $t_c = \sqrt{LC_1} = 2.35 \times 10^{-5} \text{ s}$ yields the following set of dimensionless equations:

$$\frac{dV_1}{dt} = \frac{V_1}{R_n} - g(V_1 - V_2) + q_1, \quad (12a)$$

$$\frac{dV_2}{dt} = \frac{C_1}{C_2} [g(V_1 - V_2) - I + q_2], \quad (12b)$$

$$\frac{dI}{dt} = V_2 - R_m I + q_3, \quad (12c)$$

where $g(V) = V/R_d + 2I_r \sinh(\alpha V)$, and the sans serif font indicates dimensionless quantities. Throughout the rest of the paper, we take $\mathbf{z}^T = [V_1, V_2, I]$.

Since the detailed dynamics of this chaotic oscillator have not been given elsewhere, we first characterize its behavior in the absence of control. The potentiometer R_p serves as the bifurcation parameter; increasing the resistance shunts less current to ground and hence increases the current flowing to the LRC section of the circuit through the nonlinear diodes, giving rise to more complex dynamical behavior. For small R_p , the system resides on a stable steady-state p_0 with coordinates $(0, 0, 0)$. It undergoes a pitchfork bifurcation at $R_p = R_d$, above which the system resides on a stable steady state p with coordinates $(V_1^{ss}, V_2^{ss}, I^{ss})$, or p' with coordinates $(-V_1^{ss}, -V_2^{ss}, -I^{ss})$, depending on the initial conditions. From Eq. (11), we find that V_1^{ss} is given by the roots of the transcendental equation,

$$V_1^{ss}/R_n - g[V_1^{ss}(1 - R_m/R_n)] = 0. \quad (13)$$

The other variables can be found using the relations $V_2^{ss} = R_m V_1^{ss}/R_n$ and $I^{ss} = V_1^{ss}/R_n$. At $R_p = 6.5 \text{ k}\Omega$, the system undergoes a Hopf bifurcation to a stable period-1 orbit.

For increasing R_p , the system essentially follows a period-doubling route to “single-scroll” chaos characterized by a single positive Lyapunov exponent. The period-doubling route to chaos is not exact in that the period-doubling cascade is temporarily reversed for a small range of R_p , that is, the system displays antimonotonicity.³⁰ At $R_p = 12.5 \text{ k}\Omega$ it makes a transition to “double-scroll” chaos, also characterized by a single positive Lyapunov exponent. The system continues to evolve on a double-scroll strange attractor for increasing R_p , excluding occasional periodic windows. For example, Fig. 5 shows three experimentally observed projections of the circuit’s strange attractor in the double-scroll regime with $R_p = 20.0 \text{ k}\Omega$, where the coordinate of the trajectory in phase space is recorded using a high-accuracy, simultaneous-sampling analog-to-digital converter (National Instruments AT-A2150). The unstable steady states are indicated by the points p and p' in the center of the two “scrolls,” and the point p_0 at the origin. The characteristic time scale of the chaotic fluctuations under these conditions is approximately equal to 0.2 ms , corresponding to a characteristic frequency of $3 \times 10^4 \text{ s}^{-1}$.

V. CONTROLLER DESIGN

To stabilize the chaotic electronic oscillator about one of the USS’s, the modified ETDAS controller needs to measure a voltage or current in the circuit, filter the signal in a form given by Eq. (5), amplify it, and apply the feedback signal to

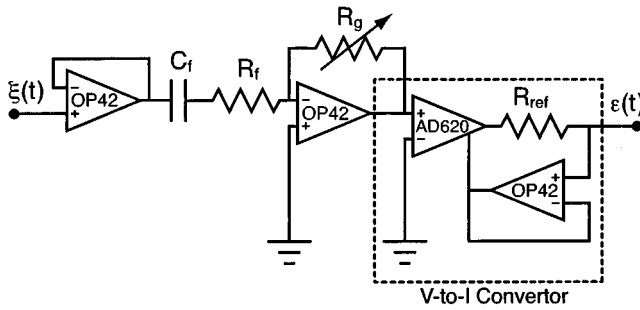


FIG. 6. Schematic for the ETDAS controller. The circuit is comprised of a buffer, high-pass filter, amplifier, and voltage-to-current converter (collection of components within the dashed box).

an appropriate accessible system parameter or variable. As a practical matter, the bandwidth of the controller must well exceed the characteristic frequency of the chaotic fluctuations ($\sim 3 \times 10^4 \text{ s}^{-1}$). A stability analysis of the dynamical system in the presence of feedback reveals that an ideal controller will stabilize the USS's p or p' when measuring any or all dynamical variables, and for feedback to a wide variety of variables or parameters. To simplify the experimental design in light of this observation, we choose to read only a single variable V_1 and apply a current proportional to the feedback signal to the V_1 -node of the circuit so that $q_1 = \bar{q}_1 + \epsilon(t)$ with

$$\frac{d\epsilon(t)}{dt} = -\omega_0\epsilon(t) - \gamma \frac{dV_1(t)}{dt}, \quad (14)$$

$\bar{q}_1 = 0$, $\hat{n}^T = [1, 0, 0]$, and $\mathbf{B}^T = [1, 0, 0]$. Measurement of other variables and feedback to different locations gives similar results.

Our controller design is shown schematically in Fig. 6. It consists of a high-impedance voltage follower that measures V_1 without disturbing the dynamical state of the circuit, followed by a single-pole high-pass filter and a voltage-to-current converter that generates the feedback signal. The voltage follower consists of a unity-gain noninverting amplifier (Analog Devices type OP-42) whose output is sent to the high-pass filter. An error signal of the form given by Eq. (14) is generated by an active high-pass filter consisting of a series-connected resistor $R_f = 5.11 \text{ k}\Omega$ and capacitor $C_f = 0.1 \text{ }\mu\text{F}$ attached to the inverting input of an operational amplifier, yielding a high-pass cut-off frequency $1/R_f C_f = 1.96 \times 10^3 \text{ s}^{-1}$ above which the controller's response levels off. The dimensionless feedback parameter appear in Eq. (14) is given in terms of this frequency through the relation $\omega_0 = t_c/R_f C_f = 0.046$. The transmission coefficient (gain) of the filter is set by the ratio R_g/R_f . Note that the high-pass cut-off frequency is well below the characteristic frequency of the chaotic fluctuations; we find that our ability to stabilize the USS's is not overly sensitive to the choice of the cut-off frequency so long as it is well below the characteristic frequency of the chaotic fluctuations. The filtered signal is converted to a current using an instrumentation amplifier (Analog Devices type AD620), a reference resistor $R_{ref} = 6.04 \text{ k}\Omega$, and a voltage follower that sets the reference

input to the instrumentation amplifier equal to the output voltage of the device. This voltage-to-current converter generates a current $I_{out} = V_{in}/R_{ref}$ while maintaining an extremely high output impedance, where V_{in} is the input voltage to the device. For future reference, the dimensionless feedback gain parameter appearing in Eq. (5) is given in terms of the component values through the relation $\gamma = R_c R_g / R_f R_{ref}$.

We find that our controller faithfully generates the desired error signal for frequencies below approximately 10^6 s^{-1} , above which the error signal becomes somewhat distorted. This nonideal behavior does not seriously affect the performance of the controller for this particular application since it accurately produces an error signal of the form given by Eq. (5) in the frequency range encompassed by the chaotic fluctuations. Note that it should be possible to stabilize ultra-high-frequency chaotic instabilities using higher speed components; high-pass filters and amplifiers are commercially available that operate at frequencies well in excess of 1 GHz. In addition, an all-optical high-pass filter can be realized approximately using a short, high-finesse Fabry-Perot interferometer.⁶

VI. CONTROLLING USS'S USING ETDAS

Our procedure for achieving control of the USS's using the modified form of ETDAS is to connect the output of the feedback circuitry to the V_1 -node of the chaotic oscillator and search for control while adjusting the feedback gain parameter γ . Since the chaotic trajectory never visits a neighborhood of the USS's p and p' (see Fig. 5), the controller must apply large perturbations to the system during the transient decay to the USS when starting from an arbitrary value of R_p . A method for controlling these USS's with small perturbations using a tracking technique^{10,31} is described below. Note that for other dynamical systems where the trajectory visits a neighborhood of the desired USS, only small perturbations would be needed to control its behavior. Successful control is indicated when the dynamical variables of the chaotic oscillator are stationary at the value of the desired USS and the feedback signal $\epsilon(t)$ drops to a very small value whose minimum size is governed by the noise level and nonideal behavior of the control circuitry [in an ideal, noise-free situation, $\epsilon(t)$ vanishes]. Our criterion for control is $|\epsilon(t)| < 5 \times 10^{-3}$.

Figure 7 shows a typical transient evolution of V_1 when control is successful to the USS p' with $R_p = 20.0 \text{ k}\Omega$ and when the control is switched on suddenly at an arbitrary instant with $\gamma = 0.34$. It is seen from the inset that the chaotic voltage decays quickly to p' . An analysis of the waveform reveals that the decay is approximately exponential with a decay time of 2.3 ms and an oscillation frequency of $2.9 \times 10^4 \text{ s}^{-1}$. Note that the initial conditions of the chaotic system at the time control is activated determines which USS the system will converge to; we find that the basin of attraction of the USS p (p') is $V_1 > 0$ ($V_1 < 0$).

For other values of the bifurcation parameter, we find that the USS's p and p' can be stabilized using a wide range of feedback gain parameters which can be visualized quickly

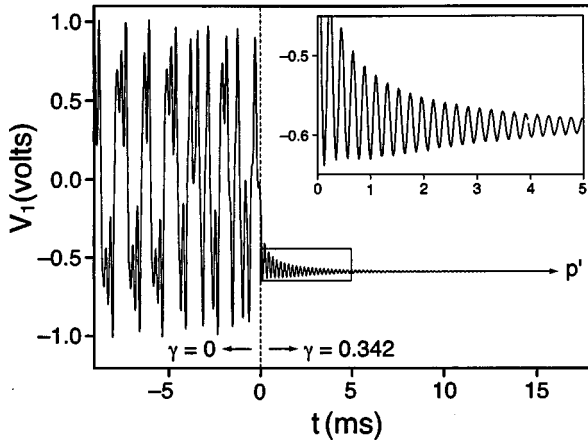


FIG. 7. Observation of control to the USS p' as indicated in the temporal evolution of the dynamical variable V_1 for $R_p = 20$ k Ω . The controller is disabled for $t < 0$, and is activated at $t = 0$ with a feedback gain $\gamma = 0.34$. The decay rate and frequency of oscillation of the waveform as the system approaches the USS agrees well with the theoretically predicted values.

from a plot of the *domain of control*. The domain of control is mapped out by determining the range of values of γ that successfully stabilize the USS as a function of the bifurcation parameter R_p . The experimentally observed boundary of the domain of control for p' is indicated by the solid line in Fig. 8 where it is seen that it is always possible to stabilize the USS's using a sufficiently large feedback gain parameter. It appears that there is no upper boundary for the domain of control since we find that control is maintained for feedback gain parameters as large as ~ 0.8 , the gain limit of our controller setup. The domain of control for the USS p is essentially identical to that for p' due to the inversion symmetry of the dynamics of the system in the presence of control.

It is of practical interest that the USS can be stabilized for any value of R_p without adjustment of the feedback parameters. As can be seen from Fig. 8, the state will always be stable for $\gamma > 0.42$. Therefore, the controller will automati-

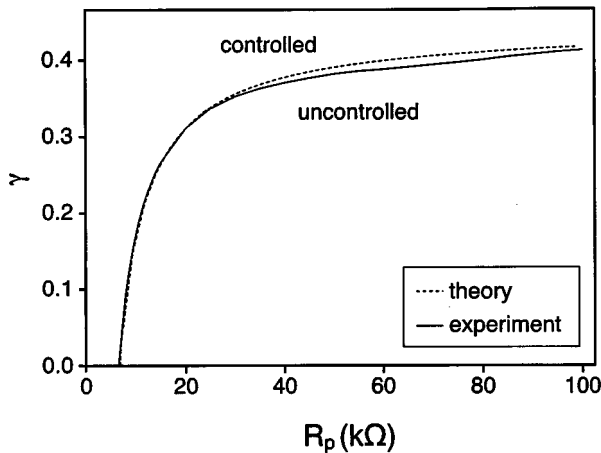


FIG. 8. Experimental (solid line) and theoretical predicted (dashed line) domains of control. The USS is stabilized by the feedback perturbations for a given value of the bifurcation parameter R_p when the feedback gain γ lies above the curve.

cally track slow changes in R_p under this condition, where the time scale of the changes must be less than $\sim 2\pi/\omega_0$ (the inverse of the corner frequency of the high-pass filter).

This property of the domain of control suggests a different method of stabilizing an USS for an arbitrary value of R_p using only small perturbations based on a tracking technique.^{10,31} The procedure is to initially adjust $R_p < 6.5$ k Ω so that the desired point (p or p') is stable in the absence of control; connect the output of the controller to the V_1 -node of the oscillator and adjust $\gamma > 0.42$; and slowly increase R_p to the desired value allowing the controller to automatically track the changes in V_1^{ss} . In practice, this procedure reliably allows stabilization of the USS's p or p' while the control perturbations never exceed 5×10^{-3} .

In contrast, we find that control of the USS p_0 is never possible for any choice of γ , $\hat{n}$, or $\mathbf{B}$, even when the system is initially placed in a neighborhood of p_0 . As will be shown in the next section, feedback of the form given by Eq. (5) fails to stabilize p_0 because it falls into the class of uncontrollable states, consistent with the discussion in the previous section and in the Appendix.

VII. THEORETICAL ANALYSIS

The goal of our controller is to stabilize the USS's p , p' , and p_0 embedded within the chaotic system as depicted in Fig. 5. Following the analysis given in Sec. III, we linearize the dynamics given by Eqs. (12) about the USS's (coordinate $\mathbf{z}_*$) with $\mathbf{q}_2 = \mathbf{q}_3 = 0$ and find that the local dynamics are governed by the matrix

$$\mathbf{A} = \begin{bmatrix} (1/R_n - g') & g' & 0 \\ (C_1/C_2)g' & -(C_1/C_2)g' & -(C_1/C_2) \\ 0 & 1 & -R_l \end{bmatrix}, \quad (15)$$

where $g' = \partial g / \partial V_1|_{\mathbf{z}_*} = -\partial g / \partial V_2|_{\mathbf{z}_*}$ and the circuit parameters are given in Sec. IV. In the presence of control with $\hat{\mathbf{n}}^T = [1, 0, 0]$, $\mathbf{B}^T = [1, 0, 0]$, the expanded-dimension dynamics including the effects of the controller are governed by

$$\mathbf{G} = \begin{bmatrix} (1/R_n - g') & g' & 0 & 1 \\ g'(C_1/C_2) & -g'(C_1/C_2) & -(C_1/C_2) & 0 \\ 0 & 1 & -R_l & 0 \\ -\gamma(1/R_n - g') & -\gamma g' & 0 & -(\omega_0 + \gamma) \end{bmatrix}. \quad (16)$$

We first consider the stability of the USS's p and p' . Under conditions when $R_p = 20$ k Ω , $g' = 6.58$ and the eigenvalues of $\mathbf{A}$ are equal to $(-12.7, 0.083 \pm i0.61)$. Hence we expect that stabilization of the USS is at least possible since there is not an odd number of real eigenvalues whose size is greater than zero. For $\gamma = 0.34$, corresponding to the conditions of Fig. 7, we find that the eigenvalues of $\mathbf{G}$ are equal to $(-12.9, -0.040, -0.0082 \pm i0.65)$. Due to the disparity in the size of the real parts of the eigenvalues, we see that the long-term decay to the USS is governed predominantly by the last eigenvalue. Therefore, the predicted exponential decay rate is $t_c/0.0082 = 2.9$ ms and oscillation frequency is

$0.65/t_c = 2.8 \times 10^4 \text{ s}^{-1}$, in reasonably good agreement with the experimentally measured waveform shown in the inset of Fig. 7.

For other values of R_p , we determine the domain of control using the Routh–Hurwitz criterion which is based on inequalities involving the coefficients of the characteristic polynomial. The characteristic polynomial of the matrix $\mathbf{G}$ is given by

$$\det[\lambda \mathbf{I} - \mathbf{G}] = \sum_{n=0}^4 \alpha_n \lambda^{4-n}. \quad (17)$$

Necessary and sufficient conditions for stability of the USS in the presence of control is $\alpha_n > 0$ for all n , and

$$\alpha_1 \alpha_2 \alpha_3 - \alpha_0 \alpha_3^2 - \alpha_1^2 \alpha_4 > 0. \quad (18)$$

Applying these conditions to our system result in the dashed line shown in Fig. 8. There is no upper boundary of the domain of control. It is seen that the agreement between our experimental observations and theoretical predictions (with no adjustable parameters) are excellent.

Finally, we consider stabilization of the state p_0 when it is unstable ($R_p > R_d$). Note that $g' = 0$ for this fixed point. We find that the matrix $\mathbf{A}$ possesses a single real positive eigenvalue and a complex conjugate pair with a negative real part. Therefore, based on our discussion in Sec. III and the Appendix, we expect that this USS cannot be stabilized using this feedback scheme, consistent with our experimental observations.

VIII. CONCLUSIONS

Our experimental observations and theoretical analysis demonstrates that continuous feedback of an error signal generated by high-pass-filtering the dynamical state of the system is well suited for stabilizing certain USS's of chaotic systems. The form of the feedback signal is motivated by an analysis of the ETDAS feedback method that has been used previously to stabilize UPO's in fast dynamical systems. The scheme is simple, does not require a reference state with which to compare the dynamics, automatically tracks slow changes in the state of the system, and is less sensitive to high-frequency noise in comparison to other methods. The next step in this work is to apply this technique to the stabilization of USS's in ultra-high-speed systems, such as electronic circuits and lasers.

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APPENDIX: CONTROLLABILITY OF CERTAIN USS'S USING ETDAS FEEDBACK

Generalizing the work of Ushio,²⁷ we prove that feedback of the form given by Eq. (5) is incapable of stabilizing

an USS characterized by a matrix $\mathbf{A}$ that has an odd number of real eigenvalues whose size is greater than zero.

Let

$$H(\lambda, \gamma) = \det[\lambda \mathbf{I} - \mathbf{G}] = \sum_{n=0}^{N+1} \alpha_n \lambda^{N+1-n}, \quad (A1)$$

where $\mathbf{G}$ is given by Eq. (10) and N is the dimension of the chaotic system in the absence of control (i.e., of the matrix $\mathbf{A}$). A necessary and sufficient condition for failure of control is to show that $H(\lambda, \gamma)$ has at least one real root greater than zero for all γ . Since $H(\lambda, \gamma)$ is a continuous function, there exists one real root whose value is greater than zero if $H(\lambda, \gamma)$ switches sign over the interval $[0, \infty)$. At one end of the interval,

$$\lim_{\lambda \rightarrow \infty} H(\lambda, \gamma) \rightarrow \infty, \quad (A2)$$

since $\alpha_N > 0$. At the other end of the interval,

$$\begin{aligned} H(0, \gamma) &= \det \begin{bmatrix} -\mathbf{A} & -\mathbf{B} \\ \gamma \hat{\mathbf{n}}^T \mathbf{A} & (\omega_0 + \gamma \hat{\mathbf{n}}^T \mathbf{B}) \end{bmatrix} \\ &= \det \begin{bmatrix} -\mathbf{A} & -\mathbf{B} \\ \mathbf{0} & \omega_0 \end{bmatrix} = \omega_0 \prod_{n=1}^N (-e_n), \end{aligned} \quad (A3)$$

where $\omega_0 > 0$, and e_n are the eigenvalues of the matrix $\mathbf{A}$. Note that this result demonstrates that $H(0, \gamma)$ is independent of γ . If $\mathbf{A}$ possesses an odd number of real eigenvalues whose size is greater than zero, $H(0, \gamma) < 0$, thus completing the proof.

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